Forms of application: (Pg 573 in Task_1_RAG_Knowledgebase)

• CHIVA 1: in only one time without creating hemodynamic affectation and obtaining a drained system. Shunt Types 1, 2 and 4. The suppression of the main leakage point can be done at the same time it acts upon the eventual N3.

• CHIVA 2: strategy in two times. Type 3 Shunts.--- In the first time it acts upon N2-N3 leakage point (the saphenous vein develops a new re-entry perforating vein on N2 transforming the Type 3 shunt into a Type 1 shunt).--- In the second time the N1-N2 leakage point is closed. It requires periodic Doppler controls and it is counterindi cated in cases where the caliber of the saphenous vein is greater than 1 cm, due to the risk of thrombosis and potential complications when the arch is left open.

• CHIVA 1 + 2: Type 3 shunts. In only one surgical act it gen erates hemodynamic conflict by compromising the system drainage; it suppresses the leakage points without orga nizing drainage. On some occasion it develops neo-N3. The intervention is usually performed on one limb only. It has a certain number of advantages compared to other more aggressive surgical techniques like stripping (removal of veins), such as a local anesthesia as opposed to rachideal anesthesia, fewer complications, immediate walking after the intervention and a faster come-back to work meaning lower socio-economic costs.15,16 There is no statistically significant difference between the healing percentages after 5 years of follow-up with both techniques.

Pg no 583 :

In case of a saphenous trunk RP absence, an interruption of the N1-N2 leaking point would lead to a not-draining venous system, because of the energy gradient suppression (see therapeutic strategy chapter). In type 2 shunt (Figure 5a), the leaking point is only from N2 to N3, with a re-entry perforator focused on the same incompetent tributary and draining toward N1. In this pattern, no recirculation is created, so establishing an ODS.

Two are the main surgical options: the so-called CHIVA 1 or CHIVA 2 procedures. CHIVA 1 is performed in case of a type 1 or 1þN3 shunt. It is addressed to the interruption of the N1-N2 compartment jump, exploiting the RP that is focused along the saphenous trunk. An eventually present incompetent tributary (type 1þN3shunt) can be flush ligated and let to be drained by its own RP. The name CHIVA 1 refers to the fact that the pro cedure is designed with both the high ligation and the eventually incompetent collaterals disconnection to be performed in one single shot (Figure 6). On the contrary, in type 3 shunts, the lacking of a valid RP focused on the saphenous trunk, prohibits the high ligation in the first step approach. In fact, per forming a CHIVA 1 procedure in a type

3 shunt would lead to a venous stasis, because of no energy gradient aspirating the saphenous blood column. The hydrostatic and lateral pressure inside the saphenous trunk would increase according to the Bernoulli Theorem, dilation would raise rather than diminish. A high risk of saphenous thrombosis would occur.

Type 1 Shunt (Type 1 - N3 shunt) ligation strategy (CHIVA 1): under local anaesthesia a high tie is per formed to treat the N1-N2 leaking point. The previously refluxing saphenous trunk is still draining in a retrograde direction also after the procedure, but into the RP focused on the saphenous trunk. At the following muscular diastole, no recirculation will occur because of the EP interruption. In case of an incompetent saphenous tributary, this will be flush ligated during the same procedure and its flow will be aspirated by the same tributary RP directly into the N1 compartment

Type 3 Shunt ligation strategy : In case of an already consistent experience in CHIVA strategy, in specific cases, it is possible to treat also a type 3 shunt by means of a single shot procedure (high ligation, flush disconnection of the incompetent N3 tributaries and removing of the competent valves in the saphenous tract that separates a valid saphenous system RP from the incompetent N3 branch where the other RP is focused on). This is CHIVA2 strategy and in CHIVA 2 first step, a simple flush ligation of the incompetent collaterals is performed. After the first step in a type3 shunt, a pathological saphenous reflux can remain into the hemodynamic project ,until it will enlarge a previously inefficient perforator focused on N2. Whenever the above quoted perforator will reach a proper calibre and aspirating property, a closed shunt will be formed among N1 and N2 networks, transforming the type3 shunt in a type1 shunt, thus allowing the second step of CHIVA2 procedure, which simply coincides with a CHIVA1 procedure (i.e. high ligation) done for a type-1 shunt. In case the saphenous reflux was fed just by a vessel dilation linked to the N3 aspiration, the N2 valves will recover their functionality, thanks to the saphenous calibre reduction, thus avoiding the necessity of a second step. Type3 shunt being transformed into a type1 shunt by the first step of a CHIVA2 procedure and then treated by a CHIVA2 second step intervention(equal to CHIVA1)

Type 2 Shunt ligation strategy : Considering the type 2 shunt compartment jump (fromN2toN3), it becomes obvious show CHIVA2 first step represents the correct indication also for this kind of reflux treatment. No further procedures are to be planned in a type2 shunt. First step of CHIVA2 procedure i.e. a flush ligation is done for type2 closed shunt.

Pg 584-587

A simple but really effective sonographic test is able to identify the presence of a valid RP focused on the saphenous vein: theRefluxEliminationTest.2 A digit compression of the incompetent tributaries will eliminate the aspiration effect provoked by the re-entry perforators focused on N3.

Reflux elimination test. A simple digit compression of the incompetent tributary that is endowed with its own RP will permit to discriminate among a type 1 (type1+N3) and a type 3 shunt. In case of saphenous reflux persistence at the digit compression, an RP must be keeping the energy gradient present, thus indicating the perforating vein along the not compressed saphenous trunk (type 1+N3 shunt). In case of reflux disappearance at the finger compression, the RP must be expected along the same N3 tributary (thus a Type 3 shunt), being excluded by the finger compression, so temporarily eliminating the energy gradient.

The Doppler must be placed cranially to the investigated saphenous RP. If the evoked saphenous reflux will be eliminated by the N3 escape points compression, the energy gradient addressed to the blood column aspiration through that RP must be considered inefficient. Consequently, it will be mandatory to avoid the high ligation in a single step procedure (CHIVA 1), since dealing with a type 3 shunt. The correct indication will be a first step of CHIVA 2 procedure.

Hybrid techniques : In specific reflux patterns, the CHIVA strategy can be also performed by means of hybrid techniques. This is the case for example of N1-N2 compartment jumps arising not from an incompetent sapheno-femoral junction but rather by an incompetent hunterian perforating vein, feeding a Type 1+N3 shunt (Figure 14). A flush ligation of the incompetent Hunterian perforating vein would require a too big skin incision nowadays. A mini-access is herein proposed, aimed to the saphenous trunk isolation at the perforating vein confluence. Once the latter is flush disconnected, the remaining perforating vein tract is easily treated by an intraluminal injection of foam sclerotherapy while twisting the same vessel. During the same procedure a flush ligation of the incompetent N3 tributary can be performed according to the traditional CHIVA strategy.

Type 1+N3 shunt, fed by an EP coming out from an incompetent hunterian perforating vein. In order to minimize the incision length, a mini skin access is performed in correspondence of the hunterian perforating vein confluence with the saphenous trunk. After its flush ligation, an intraoperative endovenous foam sclerotherapy is directly performed inside the perforating vein lumen, while twisting it. In this way, all the remaining leaking vein is treated, avoiding a deeper and wider dissection to obtain a flush ligation on the femoral vein. A traditional CHIVA strategy is then applied to treat the coexisting N3 incompetent tributary.

Recurrent N2-N3 leaking point: (a) a previously flush ligated N2-N3 leaking point recurred. The previous ligature knot (K) is clearly detectable by echo-color-Doppler, together with the saphenous trunk (GSV) and the recurrent incompetent tributary (T). (b) Colour evidence of the recurrent leaking point, together with the appearance of a perforating vein (PV) not properly draining.

CHIVA strategy in case of recurrences : A very interesting CHIVA strategy feature is the chance to treat recurrences by means of the same strategy. A recurrence can originate from a reunion of the two previously separated stumps (the saphenous trunk and the femoral/popliteal vein in case of a high ligation or of a sapheno-popliteal ligation, the saphenous trunk and its tributary in case of an incompetent N3 branch). A reflux re-appearance can come out also from a new N1-N2 or N2-N3 leaking point. In all these cases, a CHIVA strategy can be applied according to the indications that were above reported. The only technical difference in case of a high ligation in a recurrent sapheno-femoral reflux is the identification and isolation of the junction dissecting from the femoral vein downward, instead that from the saphenous trunk upward. In this way, it is possible to avoid all the post-operative scars that were created following the first high ligation, according to the procedure named Li's operation.

Sapheno-femoral and sapheno-popliteal junction competence assessment : Both in CHIVA 1 and in CHIVA 2 second step, the indication to a junction ligation is a crucial moment in which an inadequate diagnostic assessment could ruin the entire hemodynamic correction. The SFJ must be evaluated placing the sample volume on the femoral side of the terminal valve, per forming both a Valsalva (VAL) and a compression/relaxation (CR) manoeuver.7 Only in case of both the manoeuvers' positivity the junction can be considered incompetent. In the same way, the sapheno-popliteal junction must be assessed both by active (Parana` ) and passive (CR) manoeuvers, whose positivity must be contemporaneously present to diagnose a true junctional incompetence (Figure 18). In order to minimize the risk of venous stasis, the sapheno-femoral junction tributaries (superficial epi gastric vein, external pudendal vein, superficial circumflex vein) and the Giacomini's vein should be spared during the sapheno-femoral and sapheno-popliteal dis connection, respectively. Moreover, the maximum cure should be placed in performing ligations without long stumps on the femoral and popliteal side, in order to reduce the risk of recanalization and recurrence.

The ECD assessment proceeds with the B mode, color and PW mode identification of the eventually present saphenous incompetent tributaries. These last ones sometimes are easily detected even only by the B-mode because of their bigger calibre whenever compared with the saphenous trunk below their origin (Figure 19): a phenomenon that is due to a shunting effect regarding mainly the tributary itself, very probably endowed with a particularly efficient RP. Color and PW mode will determine the real eventual reflux presence into the tributary. In case of pelvic leak, the direction of a refluxing flow through the upper arch tributaries is not different from the physiological direction. Thus, it is assessed properly only whenever, conversely from the physiology, it is triggered by a Valsalva manoeuvre.

As for the N1-N2 compartment jumps, also in case of saphenous tributary ligation (N2-N3 compartment jump), maximum attention must be paid in not leaving long stumps during the disconnection, in order to diminish the recanalization risk.

Re-entry point identification : Perforating veins have always represented an intriguing topic both in the diagnosis and in the treatment of CVD.19,20 Of course, they can constitute the reflux origin (pathological perforating veins) or, conversely, a re-entry point of the reflux itself (physio logical function). Once the leaking points have been identified (N1-N2 and/or N2-N3 junctions), the RP must be found. Whenever there is a reflux, an RP must be expected: this is a consequence of the physical principle for which a flow, or a reflux, can exist only if an energetic gradient is present.2 This diagnostic step is fundamental: a mistake in the RP selection invariably leads to a not draining treated saphenous trunk, thus to a therapy failure. The RP can be focused on the saphenous trunk and/ or on the incompetent tributary. As above reported, a simple reflux elimination test is able to discriminate among a type 1 (eventually +N3) and a type 3. Subsequently, a color-mode analysis of a B-mode visible perforating vein will show a prevalent inward f low during calf muscle diastole, so indicating the efficient RP. Melting together, the ECD information concerning the deep venous system, the iliac valve, the N1-N2 junctions competence, the eventually present incompetent tributary and the RP localization will easily provide the shunt type recognition, and thus the consequent therapeutic strategy.

Conclusions : CHIVA is a surgical strategy that is aimed to suppress the reflux escape points, restoring a draining venous system, by means of selective leaking points interruptions. This saphenous-sparing procedure is minimally invasive, accessible under local anaesthesia, in an office based setting. It requires no particular devices, so rep resenting also a highly cost-effective option. The required surgical skills are minimal, as demon started by its successful use also by not-surgical physicians. To the contrary, the learning curve becomes consistent concerning the sonographic assessment, for which an adequate and meticulous training is mandatory. The consequent diagnostic effort seems to be rewarded by a significant reduction in varicose veins recurrences following a hemodynamic rather than an ablative strategy.

Pg 606-615:

Strategy : The CHIVA strategy aims at the TMP restoration, the prevention of the varicose recurrence and the preservation of the eventual venous grafts by : Gravitational hydrostatic pressure fractionation Closed and Open Deviated Shunt disconnection Veins and Re-entry Perforators preservation Open vicarious shunt preservation GSV preservation for eventual bypass need Shunts 1, 2, 4, 5, 6 are disconnected in one stage. Shunts 3 and mixed shunts are subject to peculiar strategies. In shunt 3 the intermediate N2 re-entry perforator is absent. So the simultaneous N1->N2 and N2->N3 disconnection would leave behind a thrombosed GSV trunk because no more drained. To avoid it, two strategies are proposed. The first one consists of a two steps procedure. The first step disconnects the N2->N3 Eps that restores the GSV antegrade flow65, 66 but is rarely sustainable because it leaves behind a high GHP column which causes a redo N1->N2 reflux due to secondary RP opening. So, in that case, a second step disconnects later the N1->N2 (SFJ) when the reflux recurs. The second strategy is preferable when possible, i.e. when the competent GSV trunk below the N2->N3 EP is present and not hypoplastic. It consists of the devalvulation of the competent segment of the trunk so that it allows reflux down to a lower RP perforator. Usually, this procedure is followed by a transient thrombosis and then a complete restore of the GSV trunk. In mixed shunts, the common escape point is not disconnected but only the distal segment that diverges towards the CS RP. In Shunts 4 and 5, the disconnection of the PLPs can be done without any prior pelvic endovenous procedures in the absence of true Pelvic Compression symptoms without significant recurrence.

Tactics : Most of the time the incision is cosmetic short under local anaesthesia. All sutures are non-absorbable to avoid any redo to angiogenesis caused by inflammatory consumption of the thread. The disconnections should avoid leaving behind any stump. The N2>N3 disconnection is a division-ligation of N3 flush N2 with 2 to 3 cms phlebectomy. The PLP disconnections are division ligation + fascia suture procedures. SFJ disconnection is performed flush the Common Femoral vein completed with a clip at the very SFJ. The disconnection consists of Crossotomy instead of Crossectomy because preserves the arch tributaries to prevent cavernomatous recanalization. To reduce the risk of bleeding in an ambulatory operation, a peculiar Triple Sapheno-Femoral Ligation (TSFL) was proposed68. In the case of Short Saphenous vein reflux, the SPJ disconnection is most of the time performed below the Giacomini vein junction and always in case of mixed shunt; the muscle fascia must be properly sutured with non-absorbable thread. Light compression + 12 days of preventive anticoagulation are prescribed.

Pg 693-700

CHIVA Strategies and Ligation for each Shunt Type :

Shunt 1 CHIVA strategy: Shunt 1 is N1-N2-N1. It can involve the great or small saphenous veins. One has to make sure that these shunts are not just the diastolic phase of a mixed shunt because the strategy would be different.

Shunt 1 in great saphenous vein (GSV): When the sapheno-femoral junction (SFJ) is the escape point (EP), GSV is precisely disconnected at the GSV. This disconnection could be made immediately beneath the arch in order to drain the arch and its tributaries through the SFJ. This is not because of the risk of recurrence by the arch tributaries. As a matter of fact, at this point aspiration is weak (no VMP), and terminal valve incompetence does not prevent excessive pressure in the deep network (from coughing, defecating, lifting heavy loads, etc.). For these reasons, the arch and its tributaries are better drained by distal re-entry. When the escape point is an incompetent perforator connected to the saphenous trunk, it remains preferable to disconnect the perforator itself when the corresponding VMP is not efficient. This is the case in the thigh. On the contrary, in the lower region of the leg, disconnection can be performed at the saphenous trunk immediately beneath the incompetent perforator thanks to the efficiency of calf VMP. This efficiency can be assessed by Doppler, which shows whether or not there is an inflow during VMP diastole while N2 is compressed by a tourniquet or a finger.

Shunt 1 in small saphenous vein (SSV): Usually the RP is at the saphena popliteal junction (SPJ).Theoretically, the shunting SSV has to be disconnected at SPJ. In fact, Giacomini vein (GV) hemodynamics allow disconnecting of the SSV immediately beneath its junction with GV because GV can function as a vicarious pathway in case of flow-pressure excess in the popliteal vein, thereby preventing recurrences through forced perforators of popliteal fossa.

Shunt 3 CHIVA strategy : Shunt 3 is N1-N2-N3-N1. It can involve the great or small saphenous veins. One has to make sure that these shunts are not just the diastolic phase of a mixed shunt because the strategy would be different. Shunt 3 differs from shunt 1 by N3 segment interposition between N2 and the re-entry point (REP). The strategies are also different. Actually, disconnection at the junction as in shunt 1 is problematic because it would disconnect a closed shunt at N1-N2 but leave a derived open shunt (DOS) N2-N3 called shunt 2. N3 would be less overloaded because CS shunting flow would have been interrupted and the pressure column would have been fractionated at the arch, but overload would still be excessive because of the remaining shunt 2. Disconnection at both N1-N2 and N2-N3 junctions would disconnect both shunts 1 and 2, but N2 draining flow would be precluded because there would be no available re-entry points (REP) on its track. Even if this approach is conservative, disconnects the shunts, and fractionates the hydrostatic pressure, double disconnection in shunt 3 is not CHIVA because it does not allow draining flow. In order to carry out a correct CHIVA in shunt 3, two different procedures are possible. The first is limited to one disconnection at N2-N3 that stops the shunt 1 and 2 shunt flows. In that case, N2 and N3 drain no more shunt flows. N3 is retrograde but drains only its territory, and hydrostatic pressure is fractionated at the N2-N3 junction. N2 is antegrade due to no efficient re-entry on its track, but it is still subjected to the previous nonfractioned column of hydrostatic pressure because of no N1-N2 fractioning. N2 segment submitted to excessive hydrostatic pressure explains two phenomena: 1) calibre is partially reduced due to shunt flow suppression but does not return completely to normal, and 2) in some cases a previously small and inefficient perforator located on this N2 segment can enlarge, so that N2 flow is reversed and the previous shunt 3 is converted to shunt 1. In the latter case, the induced shunt has to be corrected, like any shunt 1, by a disconnection at the N1-N2 junction. This procedure is called CHIVA 2 because it may need of two different procedures planned when the duplex investigation confirms the transformation of type 3 shunt into type 1. The latter consists of disconnections at both the N1-N2 and N2-N3 junctions associated with distal N2 devalvulation until an efficient re-entry is represented by one or more perforators.

Shunt 4 CHIVA strategy: Shunt 4 is a CS N1->N3->N2->N1. EP at the N1->N3 junction is usually a pelvic escape point, gluteal, obturatory, inguinal (I point) or perineal (P point). Disconnection must be performed only at the EP.

Shunt 5 CHIVA strategy: Shunt 5 is a CS N1-N3-N2-N3-N1. As in shunt 4, EP at the N1-N3 junction is usually a pelvic escape point, gluteal, obturatory, inguinal (I point) or perineal (P point). Disconnection must be performed not only at the N1-N3 EP, but also at the N2-N3 in order not to leave a shunt 2.

Shunt 2 CHIVA Strategy : Flush Ligation.

Shunt 6 CHIVA strategy: Shunt 6 is a CS N1->N3->N1. EP can be a pelvic escape point or any other EP. This shunt can appear in congenital malformations but is more usually seen in recurrences following removal of varices and saphenous.

Disconnection of a mixed shunt (MS): A mixed shunt is composed of two shunts: a vicarious open shunt (VOS) and a closed shunt (CS) that share the same EP and distal segments of the shunting veins. Their REP and proximal segments are different. REP of VOS is proximal to LP, and REP of CS is distal to LP. As VOS has to be preserved and CS disconnected, the disconnection cannot be at LP but only at the distal venous segment of CS where it diverges from the distal venous segment of VOS.

Disconnection of superficial derived open shunts (DOS): DOS are shunts 2, N2-N3-N1. Disconnection of DOS is done simply on N3 at the N2-N3 junction.

Disconnection of composite shunts: Practically speaking, most shunts are composite in that a shunt 1 can be connected to a shunt 2 and so on. Sometimes the REP of one shunt is not efficient enough, and drainage is also provided by the REP of an associated shunt. Therefore, any shunt can be disconnected as long as its disconnection does not preclude any draining flow in the associated shunt. For this reason, the proper REP of every associated shunt has to be assessed by dynamic tests. For example, when a composite shunt is shunt 1+2 (N1->N2->N1->N2->N3-.N1), disconnection can be carried out at N1->N2 and N3->N1 as long as REP in N2 is efficient. If REP in shunt 1 is not obviously efficient, this composite shunt can be assimilated to a shunt 3 and treated by CHIVA 2 or CHIVA devalvulation. In another case, two EPs can be superimposed and be drained by the same distal REP (N1-N2 N1-N2-N1). Fractionation can be done immediately at the upper EP. It may be also done beyond the lower EP even if refluxing during diastole, but only if testing shows it to be efficient enough as REP. Thus, under appropriate conditions, a previous EP can be transformed into REP.

Mapping is a schematic design that depicts the venous anatomo-functional configuration of each patient. Flow direction according to the systole-diastolic phase of dynamic tests and Valsalva maneuver is indicated by an arrow for each depicted vein. Data are obtained through echo-Doppler. In this way EP, shunting veins, REP, and occluded or removed veins can be identified and analyzed in order to plan an appropriate CHIVA strategy. Only echo-Doppler allows such mapping and thus a correct CHIVA strategy. Clinical examination is not sufficient to assess accurately different types of shunts. Use of an ultrasound duplex scanner is mandatory because it is the only noninvasive instrument able to show in real time the anatomy of the venous network and, contemporaneously, venous flow direction and velocity in any position and during any dynamic maneuver.

Valsalva Manoeuver (VM) Positive VM, that is an antegrade or retrograde flow elicited by thoraco-abdominal and diaphragm contraction, evidences a deep incompetence when in deep veins and a closed shunt CS when in superficial veins. Escape point is checked upstream. Valsalva maneuver is negative in VOS and DOS.

Manual or Pneumatic Squeezing of the Calf : It can evidence reflux upon relaxation but can be false negative when squeezing is performed incorrectly or when the aspiring VMP sector is not involved by compression. Moreover, squeezing compresses superficial veins, which does not occur during physiologic VMP motion.

Dynamic VMP Activation : Paranà maneuver allows a “physiologic contraction” and, at the same time, facilitates ultrasound scanning in the standing immobile position due to an isometric reflex contraction of the muscles of the lower limbs to a slight push-pull at the patient’s waist. Flow is activated by VMP systole and /or diastole.

In the superficial network, diastolic flows are elicited by the VMP in CS and DOS. Systolic flows are elicited in CS, VOS and DOS. Diastolic velocity is proportional to the shunting flow. Diastolic and systolic flow directions are usually opposite in CS and DOS but can be the same as CS that involve Giacomini vein. In the case of multiple REPs in CS or DOS, the efficiency of the REPs proximal to the terminal can be assessed by diastolic flow velocity while compressing the terminal re-entry. This assessment is useful to the hemodynamic therapeutic strategy. Reflux time, Psatakis index, or dynamic reflux index (DRI) measurement can evidence reflux only when systolic and diastolic flow directions are opposite. Flow directions remain pathologic even when pressure and velocity return to normal but remain inverted after shunt disconnection. In this case, flow direction is abnormal but no longer pathogenic. In some cases, the flow in great incompetent varicose saphena cannot be activated by diastole when the VMP is disabled because of total deep venous valve incompetence. That is why valve incompetence allows retrograde flow only if the pressure gradient is inverted, that is, when the VMP is efficient. In other cases, diastolic reflux occurs while deep venous veins are refluxing, too. This means that there remains an efficient valvulo-muscular portion connected to the shunt. In other words, the greater the superficial diastolic reflux, the better the VMP efficiency.

In the deep network, systolic and diastolic flows are elicited by the VMP. Diastolic flow evidences a proximal valve incompetence without deep CS when diastolic flow volume is less than or equal to systolic, or when DRI > 1. Deep CS is demonstrated when diastolic flow volume is greater than systolic, or DRI >1, because, as explained previously, CS carries deviated flow in addition to normal during diastole. Detection of deep CS is the central goal of deep venous incompetence investigation because it is determinant for hemodynanamic therapeutic strategy.